DEMONSTRATION OF PROPOSED FEATURES

Project Name: EduGenie

Project Type: AI-Powered Learning Assistant

Technology: Google Gemini Generative AI

1. INTRODUCTION

EduGenie is an AI-powered learning assistant developed to support students in understanding educational content.

The demonstration of proposed features shows how the main functionalities of EduGenie work and how they can help students during learning and revision.

2. OBJECTIVE OF THE DEMONSTRATION

The main objectives are:

- To demonstrate the major features of EduGenie.
- To show the interaction between the student and the application.
- To demonstrate Google Gemini AI integration.
- To show how educational content is processed.
- To demonstrate how AI-generated responses support learning.

3. FEATURE 1 – USER INPUT

Description:

The student can enter educational content, a topic, or a question into the EduGenie application.

Demonstration:

1. Open the EduGenie application.
2. Locate the input area.
3. Enter a sample educational question or content.
4. Submit the input.

Expected Result:

The application accepts the student's input and prepares it for AI processing.

4. FEATURE 2 – AI CONTENT PROCESSING

Description:

EduGenie processes the student's input before sending the request to Google Gemini.

Demonstration:

1. Enter educational content.
2. Submit the request.

3. The application prepares the input for the AI service.

Expected Result:

The input is successfully prepared and sent for AI processing.

5. FEATURE 3 – CONTENT SUMMARIZATION

Description:

EduGenie can generate a concise summary from lengthy educational content.

Demonstration:

1. Enter a passage of educational content.
2. Request a summary.
3. Submit the request.
4. Observe the generated response.

Expected Result:

A shorter version of the provided content is generated to support quick revision.

6. FEATURE 4 – SIMPLE EXPLANATION

Description:

EduGenie can provide simpler explanations of complex educational topics.

Demonstration:

1. Enter a difficult topic.
2. Request a simple explanation.
3. Submit the request.
4. View the generated explanation.

Expected Result:

The topic is explained in a clearer and easier-to-understand manner.

7. FEATURE 5 – IMPORTANT POINT EXTRACTION

Description:

EduGenie can identify important information from educational content.

Demonstration:

1. Enter study material.
2. Request the important points.
3. Submit the request.

4. View the generated key points.

Expected Result:

Important information is presented in a concise format that can help with revision.

8. FEATURE 6 – QUESTION AND ANSWER

Description:

Students can ask learning-related questions and receive AI-generated answers.

Demonstration:

1. Enter a question.
2. Submit the question.
3. Wait for the AI response.
4. Read the generated answer.

Expected Result:

EduGenie provides a relevant educational response to the question.

9. FEATURE 7 – AI RESPONSE GENERATION

Description:

Google Gemini Generative AI generates the response based on the student's request.

Demonstration:

Student Input

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EduGenie

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Gemini API

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Google Gemini AI

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Generated Response

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EduGenie

Expected Result:

The AI-generated response is returned to the application successfully.

10. FEATURE 8 – RESPONSE DISPLAY

Description:

The generated response is displayed to the student through the EduGenie interface.

Demonstration:

1. Submit an educational request.

2. Wait for the AI-generated response.

3. Observe the response displayed in the application.

Expected Result:

The response is presented clearly and can be used by the student for learning or revision.

11. FEATURE 9 – ERROR HANDLING

Description:

EduGenie handles common errors that may occur during application usage.

Examples include:

- Empty input.
- Invalid request.
- API communication failure.
- Internet connection problems.

Expected Result:

The application provides an appropriate error message instead of failing unexpectedly.

12. COMPLETE FEATURE DEMONSTRATION FLOW

The complete demonstration follows this workflow:

Student

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Enter Educational Content / Question

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Submit Request

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Input Processing

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Gemini API

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Google Gemini AI

↓

Generate Response

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Response Processing

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Display Result

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Student Uses Result for Learning

13. FUTURE PROPOSED FEATURES

Future versions of EduGenie can include:

- AI-generated quizzes.
- Flashcard generation.
- Personalized study plans.
- Learning progress tracking.
- Voice-based interaction.
- Multi-language learning support.
- Subject-specific learning assistance.

14. EXPECTED DEMONSTRATION OUTCOME

The demonstration should show that EduGenie can:

- Accept student input.
- Process educational content.
- Communicate with Google Gemini.
- Generate useful AI responses.
- Provide simplified educational information.
- Support learning and revision.
- Handle common application errors.

15. CONCLUSION

The demonstration of proposed features shows the practical functionality of EduGenie as an AI-powered learning assistant.

By combining student input processing with Google Gemini Generative AI, EduGenie provides useful educational assistance and creates a foundation for adding more advanced learning features in the future.